THE UNIVERSITY OF MELBOURNE

COMP30023 – Computer Systems

Internet (Network) Layer – Internet Control Protocols

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Recap

- Forwarding vs routing
- Flooding vs shortest path
- Bellman's optimality principle
- Dijkstra's algorithm to find the shortest path
  - Closed (decided) and open (undecided) sets
- BGP

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Summary

- Internet Control Protocols
  - ICMP
  - DHCP
  - ARP

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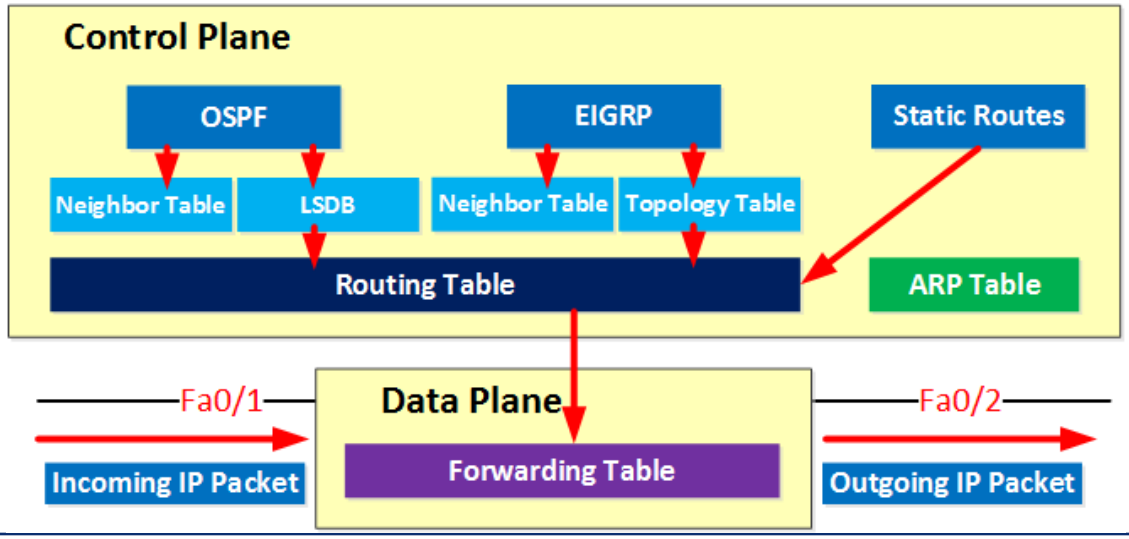
Data plane vs control plane

- Protocols don't really form a single stack
  - BGP (network layer) uses TCP (transport layer) for updates
- A better model is to think of different "planes" of protocols
  - Data plane (network layer: forwarding)
  - Control plan (network layer: choosing routes)
  - [Management plane (network layer: setting BGP policies)]



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Internet Control Protocols

- Protocols that are used at the internet layer to manage functionality
  - ICMP – Internet Control Message Protocol
  - DHCP – Dynamic Host Configuration Protocol
  - ARP – Address Resolution Protocol (not strictly internet layer)

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Internet Control Message Protocol

- You've probably all used ICMP messages (PING), but they are not limited to just sending out echo requests

Message Type	Description
Destination Unreachable	Packed could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo and echo reply	Check if a machine is alive
Timestamp request/reply	Same as Echo, but with timestamp
Router advertisement/solicitation	Find a nearby router

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Traceroute

- Exploits the Time Exceeded message
  - Recall TTL is hop count (routers the packet traverses)
  - Traceroute sends out packets to the same destination, each with an incremented TTL
  - Counters will hit zero at successive routers, causing the router to return a Time Exceeded message, revealing the IP address of the router
  - Sender can use this information to determine path and timings of the route a packet will take

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Traceroute example

- tracert on windows, traceroute on Un*x

```
tracert bbc.com

Tracing route to bbc.com [151.101.0.81]
over a maximum of 30 hops:

  1     3 ms     2 ms     5 ms  10.128.0.1
  2     4 ms     *        *     124.19.10.193
  3     6 ms     2 ms     4 ms  59.154.142.234
  4    11 ms     3 ms     8 ms  165.228.132.193
  5     6 ms     5 ms     6 ms  bundle-ether11.exi-core10.melbourne.telstra.net [203.50.11.113]
  6     5 ms     4 ms     3 ms  bundle-ether1.lon-edge901.melbourne.telstra.net [203.50.11.108]
  7     5 ms     3 ms     8 ms  fas2867546.lnk.telstra.net [110.145.207.178]
  8     5 ms     8 ms     5 ms  151.101.0.81

Trace complete.
```

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No.	Time	Source	Destination	Protocol	Length	Info
3	0.045615	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=197/50432, ttl=1 (no response found!)
4	0.046923	10.128.0.1	10.128.0.244	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
5	0.050788	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=198/50688, ttl=1 (no response found!)
6	0.052334	10.128.0.1	10.128.0.244	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
7	0.055771	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=199/50944, ttl=1 (no response found!)
8	0.057170	10.128.0.1	10.128.0.244	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
18	5.619060	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=200/51200, ttl=2 (no response found!)
19	5.624280	124.19.10.193	10.128.0.244	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
20	5.627518	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=201/51456, ttl=2 (no response found!)
21	9.389894	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=202/51712, ttl=2 (no response found!)
31	18.018353	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=203/51968, ttl=3 (no response found!)
32	18.022282	59.154.142.232	10.128.0.244	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
33	18.024359	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=204/52224, ttl=3 (no response found!)
34	18.027061	59.154.142.232	10.128.0.244	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
35	18.030934	10.128.0.244	151.101.64.81	ICMP	106	Echo (ping) request id=0x0001, seq=205/52480, ttl=3 (no response found!)

```
> Frame 3: 106 bytes on wire (848 bits), 106 bytes captured (848 bits) on interface 0
> Ethernet II, Src: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41), Dst: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
> Internet Protocol Version 4, Src: 10.128.0.244, Dst: 151.101.64.81
v Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0xf739 [correct]
    [Checksum Status: Good]
    Identifier (BE): 1 (0x0001)
    Identifier (LE): 256 (0x0100)
    Sequence number (BE): 197 (0x00c5)
    Sequence number (LE): 50432 (0xc500)
  > [No response seen]
```

```
0000  0c 8d db 8f 63 d0 ec 08  6b 1c dd 41 08 00 45 00   ....c... k..A..E.
0010  00 5c 73 06 00 00 01 01  63 71 0a 80 00 f4 97 65   .\s..... cq.....e
0020  40 51 08 00 f7 39 00 01  00 c5 00 00 00 00 00 00   @Q...9.. ........
0030  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0040  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
0050  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00   ........ ........
```

Frame (frame), 106 bytes | Packets: 131 · Displayed: 46 (35.1

Traceroute example (international)

```
Tracing route to ****************
over a maximum of 30 hops:

  1     2 ms     1 ms     4 ms  10.128.0.1
  2     4 ms     *        *     124.19.10.193
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5    16 ms    18 ms    18 ms  59.154.18.26
  6    16 ms    15 ms    16 ms  59.154.18.8
  7   169 ms   169 ms   173 ms  203.208.174.49
  8   164 ms   165 ms   165 ms  las-b24-link.telia.net [213.248.81.165]
  9   165 ms   165 ms   166 ms  las-b21-link.telia.net [62.115.136.46]
 10   167 ms   168 ms   169 ms  be3052.ccr41.lax05.atlas.cogentco.com [154.54.10.9]
 11   167 ms   173 ms   167 ms  be3243.ccr41.lax01.atlas.cogentco.com [154.54.27.117]
 12   183 ms   190 ms   180 ms  be2931.ccr31.phx01.atlas.cogentco.com [154.54.44.85]
 13   211 ms   201 ms   199 ms  be2929.ccr21.elp01.atlas.cogentco.com [154.54.42.66]
 14   200 ms   201 ms   199 ms  be2927.ccr41.iah01.atlas.cogentco.com [154.54.29.221]
 15   323 ms   324 ms   325 ms  be2687.ccr41.atl01.atlas.cogentco.com [154.54.28.69]
 16   341 ms   341 ms   341 ms  be2112.ccr41.dca01.atlas.cogentco.com [154.54.7.157]
 17   313 ms   310 ms   310 ms  be2806.ccr41.jfk02.atlas.cogentco.com [154.54.40.105]
 18   325 ms   325 ms   325 ms  be2915.ccr21.alb02.atlas.cogentco.com [154.54.40.61]
 19   327 ms   328 ms   327 ms  be2088.ccr21.ymq01.atlas.cogentco.com [154.54.43.17]
 20   310 ms   314 ms   319 ms  be3042.ccr21.lpl01.atlas.cogentco.com [154.54.44.161]
 21   339 ms   339 ms   343 ms  be2391.ccr41.lon13.atlas.cogentco.com [154.54.39.149]
 22   345 ms   343 ms   343 ms  be12497.ccr41.par01.atlas.cogentco.com [154.54.56.130]
 23   356 ms   361 ms   357 ms  be2380.rcr21.lux01.atlas.cogentco.com [154.54.78.94]
 24   322 ms   320 ms   324 ms  be3456.nr51.b038844-0.lux01.atlas.cogentco.com [154.25.12.70]
```

DHCP – Dynamic Host Configuration Protocol

- The internet layer requires each host/interface to have a unique IP address
- We could manually configure each host (certain networks do this)
  - Difficult to administer, error prone
  - Slow to respond to new devices
- DHCP is the automated way of handling IP address allocation
- Security concerns – connecting any device will issue an IP address (can apply restrictions)

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DHCP

- Network has a DHCP server for issuing IP addresses
- Host sends out a DHCP DISCOVER packet
  - Routers can be configured to relay these to the DHCP server if not directly connected to the same network
- DHCP Server receives the request and responds with a DHCP OFFER packet containing an available IP address
- IP addresses are typically issued on a lease – time after which the IP address will be reclaimed by the server and re-issued
  - Hosts can request a renewal before the lease expires
- Used to set a number of parameters
  - Default gateway, DNS servers address, time servers

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DHCP

- Leaves one obvious question…

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MAC Address

- If we don't have an IP address how does the DHCP server know where to send the response to?
  - IP runs over another network – use that network's address
- Ethernet / WiFi: use MAC (Media Access Control) Address
  - Can think of it as a globally unique identifier for the interface
  - Usually hard-coded by the manufacturer
  - Between 48 and 64 bits long
- Addressing used at the Host-to-network/data link layer
  - Sometimes called physical address, but not at physical layer
- (Not examinable)
  - Why do we need a second addressing scheme?
  - Read https://apenwarr.ca/log/?m=201708#10 for an interesting take on it

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Address Resolution Protocol

- ARP is the link between the internet layer and the underlying network layer
  - It translates an IP address into a MAC address
- Broadcasts an Ethernet (or WiFi, or…) packet asking who owns the target IP address
- Broadcast arrives at every host on the network, the owner will respond with its MAC address
- The low level sending is done via MAC addresses.
- This protocol is run a lot, even to find out how to communicate with the nearest router

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ARP in Action

22	10.011613	CiscoMer_8f:63:d0	Tp-LinkT_1c:dd:41	ARP	42 Who has 10.128.0.244? Tell 10.128.0.1
23	10.011677	Tp-LinkT_1c:dd:41	CiscoMer_8f:63:d0	ARP	42 10.128.0.244 is at ec:08:6b:1c:dd:41

```
> Frame 22: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
v Ethernet II, Src: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0), Dst: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41)
  > Destination: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41)
  > Source: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
    Type: ARP (0x0806)
v Address Resolution Protocol (request)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: request (1)
    Sender MAC address: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
    Sender IP address: 10.128.0.1
    Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)
    Target IP address: 10.128.0.244
```

Request

```
> Frame 23: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
v Ethernet II, Src: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41), Dst: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
  > Destination: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
  > Source: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41)
    Type: ARP (0x0806)
v Address Resolution Protocol (reply)
    Hardware type: Ethernet (1)
    Protocol type: IPv4 (0x0800)
    Hardware size: 6
    Protocol size: 4
    Opcode: reply (2)
    Sender MAC address: Tp-LinkT_1c:dd:41 (ec:08:6b:1c:dd:41)
    Sender IP address: 10.128.0.244
    Target MAC address: CiscoMer_8f:63:d0 (0c:8d:db:8f:63:d0)
    Target IP address: 10.128.0.1
```

Response

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ARP

- Incredibly simple
- Not particularly efficient
- Security nightmare
  - No authentication
  - Caching of responses, even when not directly requested
  - ARP spoofing is the gateway attack for most man-in-the-middle attacks
  - Provides a way of intercepting and spoofing ARP messages to associate the attackers MAC address with another hosts IP address (i.e. default gateway, DNS server, website)

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Internetworking

- We have spoken a lot about internetworking without looking at the global scale of the internet.
- The global communication we have come to take for granted are dependent on a network of undersea cables

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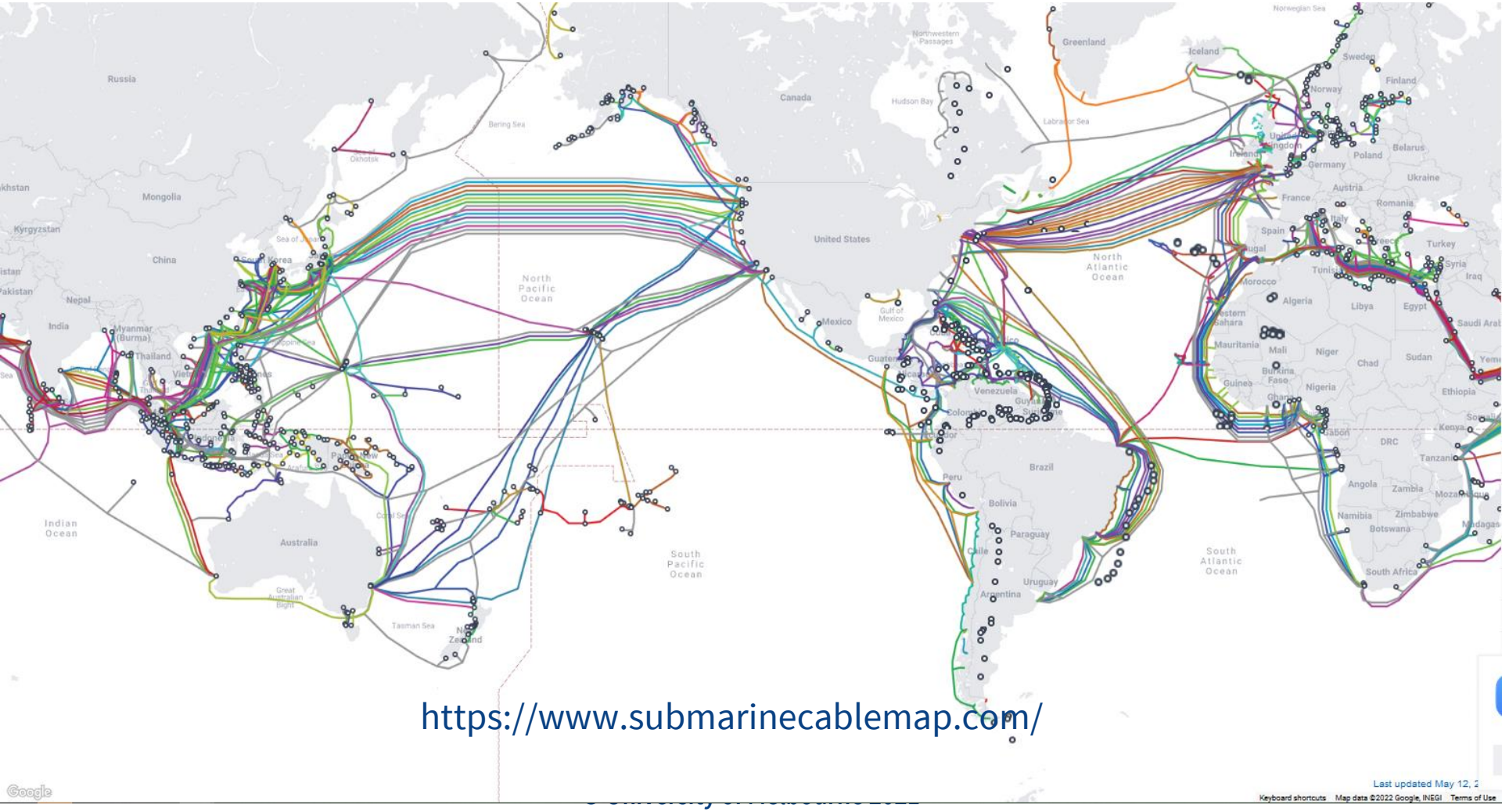

https://www.submarinecablemap.com/

And finally…

https://youtu.be/1k9-glW_5jo?t=52s

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Acknowledgement

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- Some of the images included in the notes were supplied as part of the teaching resources accompanying the text books listed in lecture 1.

- Textbook Reference: Ch 5.6, end of 4.3.3, 6.4.1

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